Session 3.4: Core Concepts of Machine Learning: Supervised vs. Unsupervised Learning and Evaluation Metrics

TODAY'S PURPOSE

Why This Session Matters

- Judge whether a model is good enough for a decision
- Look beyond a single accuracy number
- Connect model errors to business cost

Today's flow: 60 min lecture, then 120 min hands-on lab

EXPECTED OUTCOMES

By the end of today, you will be able to:

1. Compare supervised and unsupervised learning
2. Read a confusion matrix: precision, recall, and ROC-AUC
3. Pick the threshold with the lowest business cost

Concepts for This Session

CONCEPT 1

Supervised vs. Unsupervised Learning

Supervised learning trains on labeled historical targets (y), whereas unsupervised learning identifies natural groupings and structural patterns in unlabeled data (X).

CONCEPT 2

Confusion Matrix

A 2x2 contingency table tabulating classification outcomes into True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).

CONCEPT 3

Precision & Recall Trade-Off

Precision measures how many flagged cases were truly positive, while Recall measures what percentage of all actual positives were caught.

CONCEPT 4

Receiver Operating Characteristic (ROC-AUC)

A graphical plot illustrating the diagnostic ability of a binary classifier across all classification thresholds, summarized by the Area Under the Curve (AUC).

How the Concept Works in Practice

FOUNDATION 1

Supervised Target Labels vs Unsupervised Structure Discovery

When predicting fraud or customer churn, historical outcomes exist to train supervised classifiers. When discovering new customer personas or product segments, unsupervised clustering groups customers based on behavioral similarities.

FOUNDATION 2

The Asymmetric Cost of False Positives vs False Negatives

In business analytics, classification errors are rarely symmetric. In credit card fraud, a False Negative (missing a $4,000 fraudulent transaction) costs the bank $4,000. A False Positive (sending an automated verification SMS) costs $0.05. Models must be optimized for financial cost rather than naive accuracy.

FOUNDATION 3

Precision-Recall Optimization

In severely imbalanced datasets (e.g., fraud occurring in 1% of transactions), a naive model predicting "never fraud" achieves 99% accuracy but provides zero business value. Analysts evaluate Precision, Recall, and F1-Score to assess operational efficacy.

FOUNDATION 4

Cluster Inertia and Customer Archetype Discovery

In K-Means clustering, the algorithm minimizes within-cluster variance (inertia). Analysts use elbow plots to select optimal K cluster counts and label resulting clusters into actionable business segments.

Build, Test, and Diagnose in the IDE

[1] FILE EXPLORER

Organize Assets

Use the session dataset and keep scripts in a clear workspace.

[2] CODE EDITOR

Write Small Segments

Run each completed segment before expanding the program.

[3] AI AGENT CHAT

Diagnose Errors

Capture the traceback, request an explanation, and review the fix.

[4] INTEGRATED TERMINAL

Verify Execution

Confirm the final program completes with return code 0.

Script Demonstration 1: Core Concepts of Machine Learning: Supervised vs. Unsupervised Learning and Evaluation Metrics

```
import pandas as pd
    from sklearn.metrics import confusion_matrix, precision_score,
recall_score, f1_score

    df = pd.read_csv("data/session_03_04_dataset.csv")
    y_true = df["Actual_Fraud_Binary"]
    # Simulated model probability predictions
    y_pred = (df["Cardholder_Risk_Score"] >= 70).astype(int)

    cm = confusion_matrix(y_true, y_pred)
    prec = precision_score(y_true, y_pred)
    rec = recall_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred)

    print("Confusion Matrix (TN, FP / FN, TP):\\n", cm)
    print(f"Precision: {prec:.3f} | Recall: {rec:.3f} | F1-Score:
{f1:.3f}")
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 2: Core Concepts of Machine Learning: Supervised vs. Unsupervised Learning and Evaluation Metrics

```
from sklearn.metrics import roc_auc_score

    risk_prob = df["Cardholder_Risk_Score"] / 100.0
    auc_score = roc_auc_score(y_true, risk_prob)
    print(f"Model Diagnostic ROC-AUC Score: {auc_score:.4f}")
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 3: Core Concepts of Machine Learning: Supervised vs. Unsupervised Learning and Evaluation Metrics

```
from sklearn.cluster import KMeans
    from sklearn.preprocessing import StandardScaler

    features = df[["Transaction_Amount_USD", "Cardholder_Risk_Score"]]
    scaler = StandardScaler()
    scaled_features = scaler.fit_transform(features)

    kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
    df["Cluster_ID"] = kmeans.fit_predict(scaled_features)

    cluster_profiles = df.groupby("Cluster_ID")
[["Transaction_Amount_USD", "Cardholder_Risk_Score"]].mean()
    print("Unsupervised Cluster Archetype Profiles:\\n",
cluster_profiles)
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Exercise 1 and Exercise 2

EXERCISE 1: GUIDED

Foundational Practice

Follow the lab protocol, run the script, and verify its output with the session dataset.

EXERCISE 2: DEBUGGING

Applied Challenge

Extend the program, diagnose an intentional error, apply the verified correction, and confirm return code 0.